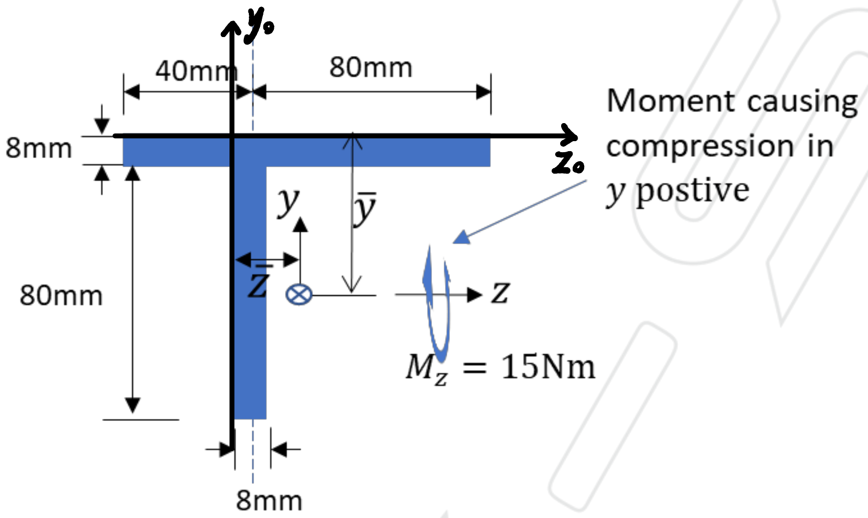
Structures I

Tutorial Sheet 5: Axial Stress in Beams

1. Calculate the maximum stress for the section below.



$$\sigma_x = -\overline{M_z}\, y$$

$$\bar{y} = \frac{4(8 \cdot 120) + 48(8 \cdot 80)}{(8 \cdot 120) + (8 \cdot 80)} = 21.6\,(mm)$$

$$\bar{z} = \frac{\overset{24}{20}(8 \cdot 120) + 4(8 \cdot 80)}{(8 \cdot 120) + (8 \cdot 80)} = \overset{16}{\cancel{13.6}}\,(mm)$$

$$\therefore\ I_z = \left[\frac{(120mm)(8mm)^3}{12} + (4mm - \overset{21.6}{\cancel{26.4}}mm)^2(8mm \cdot 120mm)\right]$$

$$+ \left[\frac{(8mm)(80mm)^3}{12} + (48mm - \overset{21.6}{\cancel{26.4}}mm)^2(8mm \cdot 80mm)\right]$$

$$= \overset{1.09}{\cancel{1.127}} \times 10^{-6}\ m^4$$

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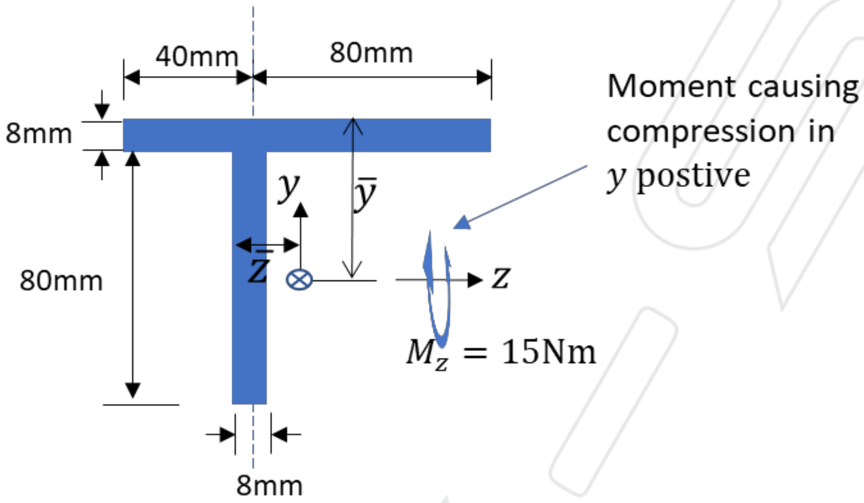

$$\therefore\ I_y = \left[\frac{(120mm)^3(8mm)}{12} + (\overset{24}{\cancel{20}}\,mm - \overset{16}{\cancel{13.6}}\,mm)^2(8mm \cdot 120mm)\right]$$

$$+\left[\frac{(8\,mm)^3(80\,mm)}{12} + (4\,mm - \overset{16}{\cancel{13.6}}\,mm)^2(8mm \cdot 80mm)\right]$$

$$= \overset{1.31}{\cancel{1.254}} \times 10^{-6}\ m^4$$

$$\therefore\ I_{zy} = 0 + (8mm \cdot 120mm)(4mm - \overset{21.6}{\cancel{26.4}}\,mm)(\overset{24}{\cancel{20}}\,mm - \overset{16}{\cancel{13.6}}\,mm)$$

$$+ (8mm \cdot 80mm)(-48mm + \overset{21.6}{\cancel{26.4}}\,mm)(4mm - \overset{16}{\cancel{13.6}}\,mm)$$

$$= -\cancel{2.703 \times 10^{-7}}\ m^4$$

$$\therefore\ \bar{M}_z = \frac{I_y M_z - I_{zy} M_y}{I_z I_y - I_{zy}^2} = \cancel{1.404 \times 10^{7}}\ N/m^3$$

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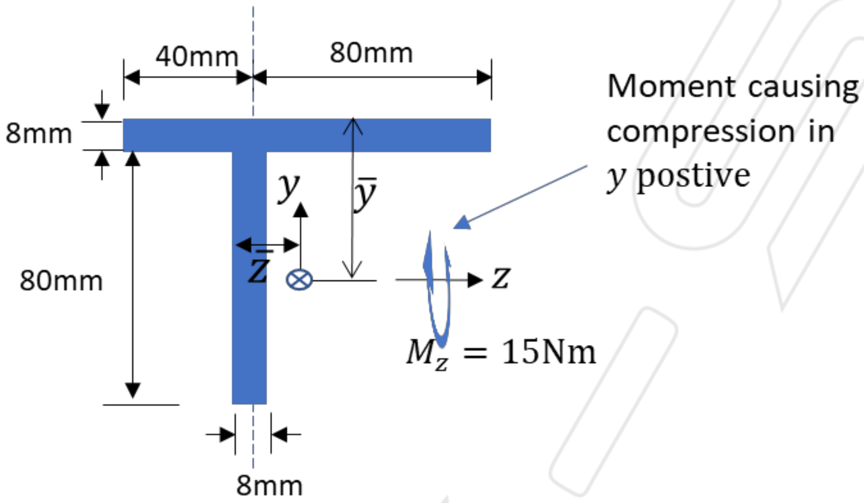


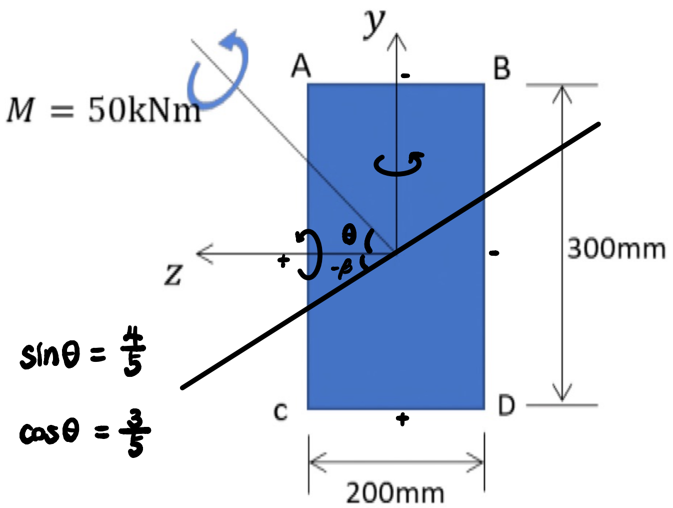
$\sigma_x = \cancel{-1.404\times10^7}\, y$

$\therefore \sigma_{x\,max}$ when $y = 21.6\text{ mm} - 88\text{ mm} = -66.4\text{ mm}$

9.6×10^5

$\therefore \sigma_{x\,max} = \cancel{9.323\times10^5}\ \text{N/m}^2$

2. Find the bending stress at the four corners of the rectangular section beam shown in figure below. Also, calculate the inclination of the neutral axis with respect to the centroidal axis.



$\sin\theta = \frac{4}{5}$

$\cos\theta = \frac{3}{5}$

$$I_y = \frac{(300mm)(200mm)^3}{12} = 2\times10^{-4}\,m^4$$

$$I_z = \frac{(200mm)(300mm)^3}{12} = 4.5\times10^{-4}\,m^4$$

$$I_{zy} = 0\,m^4$$

$$M_y = M\sin\theta = 40\,kNm$$

$$M_z = M\cos\theta = 30\,kNm$$

$$\therefore\ \bar{M}_y = \frac{I_z M_y - I_{zy} M_z}{I_z I_y - I_{zy}^2} = 2.00\times10^7\,N/m^3$$

$$\bar{M}_z = \frac{I_y M_z - I_{zy} M_y}{I_z I_y - I_{zy}^2} = 6.67\times10^7\,N/m^3$$

$$\therefore\ \beta = \tan^{-1}\left(-\frac{\bar{M}_y}{\bar{M}_z}\right) = -16.70^\circ$$

$$\therefore\ \sigma = \bar{M}_y z - \bar{M}_z y$$

$$\therefore\ \sigma_A = \cancel{8\times10^6\,N/m^2}\ 10\,MPa$$

$$\sigma_B = \cancel{-1.2\times10^7\,N/m^2}\ -30\,MPa$$

$$\sigma_C = \cancel{1.2\times10^7\,N/m^2}\ 30\,MPa$$

$$\sigma_D = \cancel{-8\times10^6\,N/m^2}\ -10\,MPa$$